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NEAR-ULTRASONIC WIRELESS ORTHOGONAL FREQUENCY DIVISION MULTIPLEXING MODEM DESIGN

- Transmit an image file through a near-ultrasonic (NUS) wireless channel.
- Investigate transmit and receive algorithms for NUS orthogonal frequency division multiplexing systems.
- Observe and analyze the waveforms and spectra of NUS systems at major processing stages.

30.1 IMAGE FILE TRANSMISSION OVER A NEAR-ULTRASONIC WIRELESS CHANNEL

In this section we transmit an image file over a near-ultrasonic (NUS) wireless channel. The image file is orthogonal frequency division multiplexing (OFDM) modulated and transmitted from the speaker of a phone over an NUS wireless channel. The microphone in a PC samples the received signal and demodulates it to restore the image data.

1.A Prepare a handheld audio device (e.g., a smartphone) that is capable of playing .wav files through its internal speaker. Also prepare a laptop or desktop PC with an internal microphone (MIC) that has installed MATLAB on it.

Step 1. [WWW]Copy the following three files into the MATLAB work folder in your laptop or desktop PC.

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Companion website: www.wiley.com/go/choi_problembasedlearning

- **NUS_AOFDM_TX.m** and **NUS_AOFDM_DEM.m** from the companion website.
- Your photo image file. If you do not have one in your PC, take one with your phone and transfer it to the MATLAB work folder via email or a USB cable.

Step 2. Change the filename of your photo to **myphoto** and maintain the file extension.

Step 3. Execute **NUS_AOFDM_TX.m**. NOTE: If your photo file extension is not `'jpg'`, then open **NUS_AOFDM_TX.m** and replace the argument `jpg` in the fourth line **'A = imread('myphoto', 'jpg')** with your photo file extension.

Step 4. Make sure that **NUS_AOFDM_TX.wav** is created in the MATLAB work folder. Move it to your phone memory or disk via email or USB cable. Do not accept any file format conversion or encoding if you are asked during the process of transferring the .wav file to your phone. For iPhones, the Gmail (Google mail) attachment and outlook attachment are two ways confirmed working.

Step 5. In the audio setting of your laptop, for recording, select the internal MIC as the default MIC. To avoid distortion in recording, disable the sound effects if there are any.

Step 6. Get ready so that you can start to run **NUS_AOFM_DEM.m** immediately and at any time needed. For example, you may type in **NUS_AOFM_DEM** in the command window but do not press the return key to run it; when you need to run this file, just need to press the return key. Alternatively, you may open **NUS_AOFM_DEM.m** and get ready to click the green play (run) button.

Step 7. Set the phone speaker volume to the maximum level and place the phone within 5 cm from the PC's internal MIC. Orient the speaker of the phone (not the receiver speaker that will be placed near your ear while you are making a phone call) toward the PC's MIC.

Step 8. Play **NUS_AOFDM_TX.wav** in the phone. After the wave file starts playing, run **NUS_AOFDM_DEM.m** which has been set ready to run in the PC in Step 6. Be sure not to stop or pause playing the wav file until **NUS_AOFDM_DEM.m** finishes running.

1.A-1 When all these steps are finished, capture the following.

(a) **Figure 1.** Be sure to maximize the window size prior to capturing. Save the **Figure 1** in .fig format, since it will be needed in Section 30.3.
(b) **Figure 3.**
(c) The value of **fo_pilot** returned in the command window. This value will be used later in 3.D-3.

1.A-2 Increase the distance between the phone and the PC bit by bit and repeat Step 7 at each distance. No need to save **Figure 1**, nor to record the value of **fo_pilot**. Record the maximum distance where the demodulated image data in **Figure 3** is recognizable.

30.2 ANALYSIS OF OFDM TRANSMITTER ALGORITHMS AND THE TRANSMITTED SIGNALS

2.A In the part **'OFDM signal parameter'** of **NUS_AOFDM_TX.m**, all system parameters to generate the NUS OFDM signal are set and their definitions are explained using comments.

2.A-1 Tabulate the parameter names, definitions, and values.

2.A-2 From the values tabulated in 2.A-1, calculate the following.

(a) Subcarrier frequency spacing in [Hz].
(b) The total signal bandwidth in [Hz], that is, the highest subcarrier frequency minus the lowest subcarrier frequency.
(c) The guard interval (cyclic prefix) length in [seconds].

2.B In the part **'OFDM modulated image data packet generation'** of the file **NUS_AOFDM_TX.m**, the image data are represented by the elements of the vector **Pixel**. This data stream is then OFDM modulated, and the complex baseband OFDM signal is stored in vector **x**.

In this part of the m-file, in the inner **'for'** loop, a certain number of image pixels are modulated into one OFDM symbol. Then a certain number of ODFM symbols are concatenated to generate one OFDM frame. In the outer **'for'** loop, the multiple OFDM frames for the entire image are concatenated to form one transmission packet. That is, **k** in the outer **'for'** loop is the OFDM frame index in the transmission packet and `kk` is the OFDM symbol index in each OFDM frame.

2.B-1 Fig. 30.1 shows the data packet structure as a frequency (subcarrier)–time (OFDM symbol) matrix for the transmission of one image. Each cell denotes one subcarrier of an OFDM symbol. From the definition of parameters listed in 2.A-1 and the part **'OFDM modulated image data packet generation'** of the m-file, determine the values of the three variables, A, B, and C in the diagram.

2.B-2 In part **'OFDM modulated image data packet generation'** of the m-file, the input to the IFFT for generating an OFDM symbol is contained in the vector **D**. According to the m-file, a certain subcarrier is allocated as the pilot subcarrier. (a) Determine the pilot subcarrier index. (b) Shade all the cells for the pilot subcarriers in Fig. 30.1.

2.B-3 The outer and inner **'for'** loops show that each OFDM frame has one fixed OFDM symbol, which is called the "pilot OFDM symbol." The pilot OFDM symbol is known to the receiver for frame synchronization and the channel estimation, whereas the pilot subcarrier described in 2.B-2 is used for carrier recovery in the receiver. Analyze the m-file, determine the OFDM symbol position in each OFDM frame, and mark all shaded OFDM symbols in Fig. 30.1.

2.B-4 Examine the m-file and determine (a) How many image data symbols (elements of the vector **Pixel**) are carried by one subcarrier of one OFDM symbol? (b) If

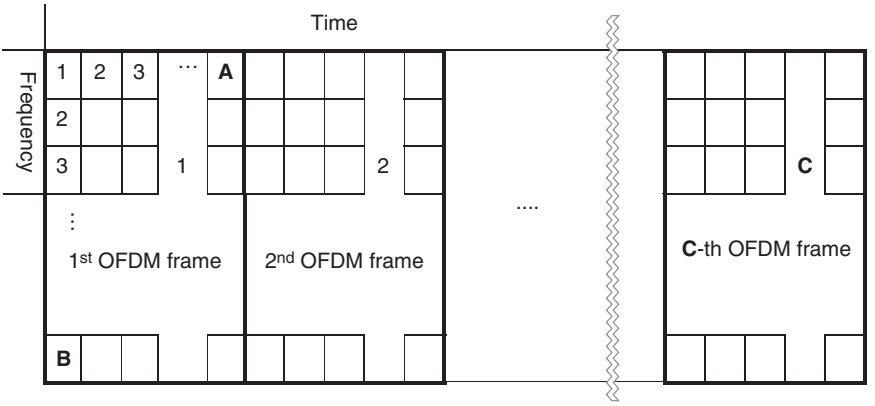


FIGURE 30.1 Packet structure to transmit one image.

one subcarrier carries more than one image data symbol, explain how it is possible to transmit multiple symbols on a single subcarrier.

2.C In part **'up conversion'** of the file **NUS_AOFDM_TX.m**, the complex baseband signal vector **x** is first up-converted into NUS frequencies. Then the sampled vector of the real-valued passband OFDM signal `tx` is generated.

2.C-1 After up-conversion, the frequency of the first (leftmost) subcarrier equals **fc**. Explain this on the basis of the m-file.

2.C-2 Based on the pilot subcarrier location determined in 2.B-2(a) and the frequency of the first subcarrier, determine the pilot subcarrier frequency in the passband.

2.D Execute **NUS_AOFDM_TX.m** and capture **Figure 2**.

2.D-1 According to the m-file, what are plotted in **Figure 2**?

2.D-2 Based on the plots in **Figure 2**, summarize all evidences observed that verify that the OFDM signal is generated correctly in the NUS band.

2.D-3 In the top subplot of **Figure 2**, zoom in the area of the OFDM spectrum and capture it.

2.D-4 From the figure in 2.D-3, (a) measure the bandwidth of the OFDM signal and record it. (b) Is the measured bandwidth consistent with the answer to 2.A-2(b)? If not, revisit 2.A-2(b).

2.D-5 From the captured figure in 2.D-3, (a) measure the pilot subcarrier frequency and record it. (b) Is the measured bandwidth consistent with the answer to 2.C-2? If not, revisit 2.C-2.

2.E In the first line of the part **'Packet repetition and side lobe suppression'** of the m-file, OFDM modulated image data packet **tx** is repeated one more time in order to

avoid a packet loss in the receiver even if we start executing the m-file late after the .wav file is being played.

In the second line, the side lobes (spectrum skirt) of the OFDM signal are suppressed by passing the OFDM signal through a bandpass filter. To check the necessity of this bandpass filtering, comment out the second line **'tx=conv(tx,bpf)'** and run **NUS_AOFDM_TX.m** again.

2.E-1 (a) Capture **Figure 2**. (b) Compare the captured figure with the one in 2.D. Assess the difference of the OFDM spectra with and without bandpass filtering.

2.E-2 Execute **soundsc(tx,fs)** in the command line. You may or may not hear any sound. Explain why it should be the case.

2.E-3 Based on 2.E-1 and 2.E-2, document why bandpass filtering for the OFDM signal is required prior to it being transmitted.

2.E-4 Uncomment the line **'tx=conv(tx,bpf)'** and run **NUS_AOFDM_TX.m** again. Then repeat 2.E-2.

30.3 ANALYSIS OF OFDM RECEIVER ALGORITHMS AND THE RECEIVED SIGNALS

3.A In the part **'Near ultrasonic sound sampling'** of the file **NUS_AOFDM_DEM.m**, the received signal by the MIC in the PC is sampled and converted into a MATLAB vector.

3.A-1 Open **Figure 1** saved in 1.A-1. According to the m-file, what are plotted in the first and second subplots of **Figure 1**?

3.A-2 Compare the first and second subplots with the captured plots in 2.D. (a) Document the main differences (b). Explain what causes such differences.

3.B In the part **'Front end BPF stage'** of the file **NUS_AOFDM_DEM.m**, the OFDM spectrum located in the NUS band is extracted by bandbass filtering.

3.B-1 According to the m-file, what are plotted in the third and fourth subplots of **Figure 1**?

3.B-2 In **Figure 1**, compare the first subplot with the third one; also compare the second subplot with the fourth one. Document the reasons that bandpass filtering is needed at this stage.

3.B-3 For all the lines in the part **'Front end BPF stage'**, explain what the variable of the left-hand side represents and how the right-hand side expression is formulated accordingly.

3.B-4 In the fourth subplot, zoom in the area of OFDM spectrum and capture it.

3.B-5 According to the captured spectrum in 3.B-4, determine whether or not the wireless channel is frequency selective for the signal being transmitted. Justify your answer.

3.C In the part **'Down conversion'** of the file **NUS_AOFDM_DEM.m**, the output of the BPF is down-converted into the baseband. The vector **cbb** is the sampled version of the complex baseband signal.

For all the lines in the part **'Down conversion'**, explain what the variable of the left-hand side represents and how the right-hand side expression is formulated accordingly.

3.D In the part **'Carrier recovery'** of the file **NUS_AOFDM_DEM.m**, the residual frequency offset in the complex baseband signal after down-conversion is measured and compensated. The vector **rcbb** is the sampled version of the frequency-offset-compensated complex baseband signal.

Since a carrier signal of the same frequency **fc** is required for up-conversion in the TX and for down-conversion in the RX (see the part **'Down conversion'**), ideally there should not be a carrier frequency offset in the down-converted complex baseband signal. In practice, however, the D/A converter clock for the phone audio device that plays the .wav file and the A/D converter clock for the PC audio device cannot be identical. This results in a residual frequency error in the down-converted complex baseband signal vector `cbb`, which should be compensated for before demodulation.

If the residual frequency offset is zero, then the pilot subcarrier frequency in the complex baseband signal `rcbb` is zero, and thus the pilot subcarrier spectrum is located at zero frequency. On the contrary, if the frequency offset is not zero, then the pilot subcarrier spectrum is located at the frequency offset. Thus, in the part `'Carrier recovery'` of `NUS_AOFDM_DEM`.m, the pilot subcarrier in `cbb` is extracted by a narrowband low pass filter, and then its center frequency, which is equal to the residual frequency offset, is measured by using FFT. In the first three lines, `'cbb'` is down-sampled 100:1 to lower the computational complexity of FFT to a practically reasonable level.

3.D-1 For the fourth line to the last line of the part **'Carrier recovery'**, explain what the variable of the left-hand side represents and how the right-hand side expression is formulated accordingly.

3.D-2 The fifth subplot in **Figure 1** shows the spectrum of the pilot subcarrier extracted from `rcbb` by using a low pass filter. Zoom in the spectrum peak to measure the residual frequency error and record it.

3.D-3 The value of **fo_pilot** recorded in 1.A-1(c) is the estimated residual frequency error. Is the measured frequency error in 3.D-2 approximately equal to **fo_pilot**?

3.E To demodulate each OFDM symbol, we need to separately perform FFT on the properly partitioned portions of **rcbb**. Proper partitioning of **rcbb** requires locating the starting sample of each OFDM symbol. The part **'Frame synchronization'** in **NUS_AOFDM_DEM.m** locates the starting point of the frame first. Then the starting point of each OFDM symbols is determined by using the frame starting point.

First, the complex baseband signal vector **rcbb** passes through the matched filter, which is matched to the pilot OFDM symbol. The filter output is then partitioned into parts, each with a length equal to OFDM frame length. After this, the parts are aligned up in parallel and summed up. Because the same pilot OFDM symbol is repeatedly and periodically inserted in every OFDM frame, the pilot OFDM symbols at the matched filter output throughout the packet are coherently added up while the data OFDM symbols are added up randomly.

Consequently, as the accumulation goes on, the peak value of the matched filter filtered pilot OFDM symbol gets significantly larger than the off-peak value in the accumulator output. The starting sample of the pilot OFDM symbol is one OFDM symbol ahead of the peak sample point of the filtered (by the matched filter) pilot OFDM symbol.

In the first three lines, in order to lower the computational complexities of frame synchronization, the complex baseband signal sample vector **rcbb** is down-sampled to generate its down-sampled version **rcc_dec**.

3.E-1 From the fifth line to the last line of the part **'Frame synchronization'**, the output of the matched filter, which is matched to the pilot OFDM symbol, is partitioned into parts, each with a length equal to the frame length and then accumulated to find the peak point. The starting sample index of the pilot OFDM symbol is then determined and stored in the variable **ST**.

Document what the variable on the left-hand side of the equal sign represents and how the right-hand side expression is properly formulated accordingly.

3.E-2 The fourth line **'rcbb_dec=abs(conv(rcbb_dec, conj(Pf_GI_dec(end:-1:1))));'** realizes the filtering process by a filter whose impulse response is equal to **'conj(Pf_GI_dec(end:-1:1))'** and then takes the absolute value of the filter output.

(a) The variable **Pf_GI_dec** is the down-sampled version of the vector **Pf_GI** that is defined in the transmitter m-file **NUS_AOFDM_TX.m**. The impulse response of the filter **'conj(Pf_GI_dec(end:-1:1))'** is the conjugate and time-reflected (left-right) version of **Pf_GI_dec**.

Explain what the filtering process in the fifth line is supposed to do.

(b) Based on the purpose of the filtering, explain why the impulse response of the filter is set as **'conj(Pf_GI_dec(end:-1:1))'**, that is, why **Pf_GI_dec** is time-reflected and conjugated.

(c) Assuming imperfect carrier recovery, explain why **abs( )** is performed after this filtering process.

3.E-3 The sixth subplot of **Figure 1** shows the filtered and accumulated OFDM frame. Is this result shown in this plot what you expected to see? Justify the answer.

3.F In the part **'OFDM demodulation with Channel compensation'**, the OFDM symbols in the packet are demodulated one by one by using FFT. The channel gains at each subcarrier are estimated by using the FFT output of the periodically inserted

pilot OFDM symbols. The FFT output vector of the data OFDM symbols is compensated for by using the estimated channel fading gains. Finally, the image data stream `Pixel` is recovered.

3.F-1 The OFDM symbols are demodulated through the outer **'for'** loop one by one by increasing the OFDM symbol index **k**. (a) In the **'for'** loop, the first line **'T=fft(rcbb(ST+GI+ NpGI*(k-1)+(1:Np)));'** performs FFT of the up-sampled vector of the current (**k**-th) OFDM symbol. Explain why the index of the vector **rcbb** is set as **'ST+GI+ NpGI*(k-1)+(1:Np)'**.

(b) The second line **'D=T(1:N);'** extracts the first **N** samples of the FFT output to get the modulated symbol vector at **N** subcarriers. Explain why **D**, that is, the first **N** samples of the FFT output, corresponds to the modulated symbol vector at **N** subcarriers.

3.F-2 Inside the outer **'for'** loop, the line **'if mod(k,Ns)==1'** checks whether the current (**k**-th) OFDM symbol is pilot or data. Explain how it does this.

3.F-3 For the two lines **'ch=D./P;'** and **'ch_tbl(frcnt,:)=ch;'** below the line **'if mod(k,Ns)==1'**, document what the variable on the left-hand side of the equal sign represents and why it is set or computed as the right-hand side. The index **frcnt** is the frame counter. The vector **P** is defined in the part **'OFDM signal parameter'** of **NUS_AOFDM_TX.m**. Section 2.B of Chapter 27 will help answer this question.

3.F-4 In the line **'D_buffer=[D_buffer ; D];'** below **'else'**, the data symbol vector **D** for the current OFDM symbol in the current OFDM frame is accumulated and stored as an additional row of the matrix **D_buffer**. If the condition checking **'if mod(k,Ns)==1'** returns 'true' after each iteration, then a new frame will start. Thus, at this point, all the data symbol vectors of the current frame have been stored in **D_buffer**. Then the innermost **'for'** loop performs channel compensation for the data symbols in the current OFDM frame in **D_buffer** to generate the final demodulation values of the data symbols and recover the **Pixel** stream in the current OFDM frame.

According to the frame structure investigated in 2.B-3, the data OFDM symbols in each frame are packed between two pilot OFDM symbols of two adjacent frames. Inside the innermost **'for'** loop, `Pch` and `Cch` are the estimated channel gain vectors from the pilot OFDM symbols of the current frame and the next frame, respectively. The channel gain vectors for the data OFDM symbols in the middle are estimated through linear interpolation using the two end values, that is, **Pch** and **Cch**.

For the three lines inside the innermost '**for**' loop, document what the variable on the left-hand side of the equal sign represents and why it is set or computed as the right-hand side.

3.F-5 Recall that the fourth subplot in **Figure 1** shows the received signal spectrum. Zoom into its spectrum portion. The seventh subplot shows the 3-D surface of the estimated channel gain magnitude over the time and frequency matrix. Compare these two subplots. Does the estimated channel gain magnitude match well the actual one? Justify your answer.

3.F-6 Based on the seventh subplot in **Figure 1**, is the current wireless channel more selective in time or in frequency? Justify your answer.

3.G The last part **'Image packet synchronization'** first locates the start of the image data packet from the demodulated pixel steam **DemPixel**. Then the vector **DemPixel** is cyclically shifted so that the first element of it becomes the beginning of the packet.

3.G-1 For all lines in the part **'Image packet synchronization'**, document what the variable on the left-hand side of the equal sign represents and why it should be set or computed as the right-hand side.

3.G-2 What does the ninth subplot in **Figure 1** show? Is it what you expected to see?

30.4 EFFECTS OF SYSTEM PARAMETERS ON THE PERFORMANCE

4.A Insert the following two lines right after the line **'RX_time=TX_time+1;'** at the beginning part of **NUS_AOFM_DEM.m** and save the m-file.

- **load NUS_AOFDM_TX_tx.mat;**
- **soundsc(tx,fs)**

Now, running **NUS_AOFM_DEM.m** will transmit the `NUS OFDM` signal from the PC's speaker, which is then received and demodulated. Therefore, without playing the .wav file in the phone, transmission and reception are executed within the m-file.

4.A-1 Maximize the PC audio volume and then run **NUS_AOFM_DEM.m**. Capture the received image in **Figure 3**.

4.A-2 Open **NUS_AOFDM_TX.m**. and change the line **'GI=ceil(Np*1/8);'** into **'GI=0'**. Run **NUS_AOFDM_TX.m** and then run **NUS_AOFDM_DEM.m**. Capture the received image in **Figure 3**.

4.A-3 Compare the quality between the received images in 4.A-1 and 4.A-2. Justify the comparison result.

4.B Answer the following questions:

4.B-1 In **NUS_AOFDM_TX.m**, **Np** is set proportional to **N**. This maintains a constant total data speed regardless of **N**. Prove this assuming **GI** $= 0$.

4.B-2 Recall that the total bandwidth has been calculated from **N**, **Np**, and **fs**. Show that with **Np** proportional to **N**, the total bandwidth is also constant, regardless of **N**.

4.B-3 Run **NUS_AOFDM_TX.m** with **N** $= 16$ and 256 and capture the received images in **Figure 3** for both cases.

4.B-4 Compare the transmitted signal waveforms in terms of the peak amplitude for **N** = 16, 64 (captured in 2.D), and 256. Which case has the highest peak power and why?

4.C Restore **N** = 64 in **NUS_AOFDM_TX.m**. The line **'Pixel=Pixel(ILindex)'** in **NUS_AOFDM_TX.m** interleaves the image pixel elements in **Pixel**. Comment out this line and run **NUS_AOFDM_TX.m**.

4.C-1 Capture **Figure 2**. Compare the transmitted signal waveform with the one captured in 2.D. Which one has a higher peak?

4.C-2 Considering the typical characteristics of adjacent pixels of a photo, explain why the transmitted waveform has a high peak power if the pixel stream is only OFDM-modulated as is.

4.C-3 Explain why interleaving of the pixel elements decreases the peak power of the OFDM modulated waveform.

4.D The algorithms and the parameter settings in the two m-files provided could be improved in terms of performance and complexity.

4.D-1 Revise the two m-files to improve the received image quality. (a) Explain where and why you revise so. (b) Capture the received images before and after the revision and confirm that the image quality did improve after the revision. For fair comparison, be sure to maintain the same signal bandwidth and the same channel environment, for example, noise level and audio volume.

4.D-2 Revise the two m-files to achieve a higher spectral efficiency, for example, a lower bandwidth while the packet transmission time is not changed, or a shorter packet transmission time while the bandwidth is not changed. (a) Explain where and why you revise so. (b) Capture the results that show a higher spectral efficiency, for example, smaller bandwidth or shorter transmission time. (c) Capture the received images before and after the revision to confirm that the image quality is not reduced due to the revision.

4.D-3 Revise the two m-files to reduce the computational complexity without lowering the received image quality. (a) Explain where and why you revise so. (b) Calculate the computation reduction ratio for the algorithm part revised. (c) Capture the received images before and after the revision to confirm that the image quality is not reduced due to the revision.